Introduction to Derivatives

> **Module:** Derivatives Basics
> **Chapter:** 01 - Introduction to Derivatives
> **Level:** Beginner

Learning Objectives

After completing this chapter, you will be able to:

- Define a derivative.
- Understand why derivatives exist.
- Differentiate between the underlying asset and the derivative.
- Identify the major types of derivatives.
- Explain how derivatives are used in financial markets.
- Understand the benefits and risks of derivatives.

What is a Derivative?

A **derivative** is a financial contract whose value is **derived from (depends on)** the value of another asset.

The original asset is called the **Underlying Asset**.

A derivative itself has **no independent value**.

Its price changes whenever the price of the underlying asset changes.

Simple Definition

> A derivative is a contract between two or more parties whose value depends on an underlying asset.

What is an Underlying Asset?

An **underlying asset** is the asset from which a derivative gets its value.

Examples include:

Asset Type	Examples
Stocks	Reliance, TCS, Infosys

Asset Type	Examples
Indices	Nifty 50, Sensex
Commodities	Gold, Silver, Crude Oil
Currency	USD/INR, EUR/USD
Interest Rates	Government Bonds
Agricultural Products	Wheat, Cotton

Basic Idea

Imagine wheat costs **₹2,000 per quintal** today.

A bakery worries that wheat prices may rise next month.

The bakery signs an agreement today to buy wheat after one month at **₹2,050**.

This agreement is **not wheat itself**.

It is a **contract based on wheat**.

That contract is a **derivative**.

How Derivatives Work

Underlying Asset → Price Changes → Derivative Contract → Profit or Loss

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Whenever the price of the underlying asset changes, the derivative's value also
changes.

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# Real-Life Example

Suppose:

- Current Gold Price = ₹60,000 per 10g
- You believe prices will rise.

Instead of buying physical gold, you purchase a **Gold Futures Contract**.

### Scenario 1
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Gold rises to ₹63,000.

Your futures contract also increases in value.

Profit ✓

### Scenario 2

Gold falls to ₹58,000.

Your contract loses value.

Loss ✗

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# Why Were Derivatives Created?

Originally, derivatives were created to reduce business risk.

Farmers wanted protection against falling prices.

Manufacturers wanted protection against rising prices.

Importers wanted protection against currency fluctuations.

Exporters wanted stable exchange rates.

Over time, derivatives also became investment and trading instruments.

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# Major Uses of Derivatives

## 1. Hedging

Protect against future price movements.

Example:

- Airline hedging fuel prices
- Exporter hedging USD/INR rate

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## 2. Speculation

Attempt to earn profit by predicting price movement.

Example:

Trader believes Nifty will rise and buys Nifty Futures.
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## 3. Arbitrage

Profit from price differences in different markets.

Example:

Gold trades cheaper in one exchange than another.

Professional traders exploit the difference.

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## 4. Portfolio Management

Large investors use derivatives to reduce portfolio risk.

Example:

Mutual funds hedge market risk using index futures.

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# Types of Derivatives

There are four major derivative contracts.

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Derivatives
│
├── Forward
├── Futures
├── Options
└── Swaps
````

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1. Forward Contract

- Private agreement
- Customized
- Not traded on exchanges
- Higher counterparty risk

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2. Futures Contract

- Standardized agreement
- Traded on exchanges
- Highly regulated
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- Daily settlement

Examples:

- Nifty Futures
- Gold Futures
- Crude Oil Futures

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3. Options Contract

Provides the **right**, but **not the obligation**, to buy or sell an asset.

Two types:

- Call Option
- Put Option

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4. Swaps

Agreement to exchange cash flows.

Common examples:

- Interest Rate Swap
- Currency Swap

Mostly used by banks and large corporations.

Exchange-Traded vs OTC Derivatives

Feature	Exchange Traded	OTC
Regulation	High	Low
Standardized	Yes	No
Counterparty Risk	Low	High
Transparency	High	Low
Liquidity	High	Lower

Examples:

Exchange:

- NSE
- BSE
- CME

OTC:

- Banks
- Financial Institutions

Participants in Derivatives Market

Participant	Purpose
Hedgers	Reduce Risk
Speculators	Earn Profit
Arbitrageurs	Price Difference Profit
Market Makers	Provide Liquidity

Advantages of Derivatives

- Risk management
- Price discovery
- High liquidity
- Portfolio diversification
- Leverage
- Lower transaction costs
- Efficient capital usage

Risks of Derivatives

- High leverage can magnify losses.
- Market volatility.
- Counterparty risk (OTC).
- Liquidity risk.
- Complexity.
- Regulatory risk.

Advantages vs Risks

Advantages	Risks
Hedging	Leverage Risk

Advantages	Risks
Liquidity	Volatility
Capital Efficiency	Complexity
Price Discovery	Counterparty Risk
Diversification	Unlimited Loss (some strategies)

Common Examples

Derivative	Underlying
Nifty Futures	Nifty 50
Gold Futures	Gold
USDINR Futures	USD/INR
Reliance Option	Reliance Stock
Crude Oil Futures	Crude Oil

Key Terminology

Term	Meaning
Derivative	Contract whose value depends on another asset
Underlying Asset	Asset on which derivative is based
Futures	Standardized exchange-traded contract
Forward	Private customized contract
Option	Right, not obligation
Swap	Exchange of future cash flows
Hedging	Reducing risk
Speculation	Taking risk for profit
Arbitrage	Profit from price differences

Summary

- A derivative derives its value from another asset.
- The underlying asset determines the derivative's price.

- The four major derivative contracts are:
  - Forward
  - Futures
  - Options
  - Swaps
- Derivatives are mainly used for:
  - Hedging
  - Speculation
  - Arbitrage
- They are powerful financial tools but involve significant risk if not used properly.

Quick Revision

✔ Derivative = Contract

✔ Underlying = Original Asset

✔ Futures = Obligation

✔ Options = Right, not obligation

✔ Swaps = Exchange of cash flows

✔ Hedging = Reduce risk

✔ Speculation = Profit from price movement

✔ Arbitrage = Profit from price differences

What's Next?

➡ **02_history_of_derivatives.md**

In the next chapter, you'll learn:

- How derivatives originated.
- Evolution from agricultural markets to modern financial exchanges.
- Growth of derivatives in India and globally.
- Role of NSE, BSE, CME, and other major exchanges.